

GENERAL SPECIFICATION G4.9

Safety Equipment

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G4.9 SAFETY EQUIPMENT

G4.9.1 Safety Showers

- (a) Safety showers shall be purpose designed proprietary units suitable for indoor or outdoor use as specified.
- (b) Safety showers shall comply with ANSI Z358.1.
- (c) The water supply to the safety shower shall be tepid, with a temperature between 15 and 35°C. The Contractor shall provide insulation to the water supply pipework to maintain the supply water within this temperature range.
- (d) The showers shall incorporate nozzles specially designed to create a deluge of water overhead for rapid decontamination and an additional spray at approximately waist height to direct water onto the lower body and legs. The minimum continuous water supply duration is 20 minutes. The water supply valves shall be mechanically operated by means of a grab handle.
- (e) The showers shall be connected to a secure potable water supply to ensure highest possible availability. A pressure gauge shall be provided for the supply to each unit. The shower collecting tray shall be connected to a discharge drain pipeline discharging to a nearby drain as approved by the S.O..
- (f) The appropriate safe condition signs shall be fixed in conspicuous places. Each safety shower station shall have a prominent lighted 'Safety Shower' signage installed above the station. The fonts size of the signage shall be clearly visible from a distance of 6 m.
- (g) Each safety shower shall also be integrated with a drench hose.

G4.9.2 Eye Baths

- (a) Eye baths shall be either self-contained units or integrated with safety showers as appropriate. They shall be of a purpose designed proprietary type suitable for indoor or outdoor use as specified.
- (b) Eye baths shall comply with ANSI Z358.1.
- (c) The water supply to the eye bath shall be tepid, with a temperature between 15 and 35°C. The Contractor shall provide insulation to the water supply pipework to maintain the supply water within this temperature range.
- (d) They shall incorporate fine mesh filters and pressure regulators to ensure a safe flow of water for eye irrigation. The water supply valve shall be mechanically operated by means of a tread bar.
- (e) The eye baths shall be connected to a secure water supply to ensure highest possible availability. The minimum continuous water supply duration is 20 minutes. A pressure gauge shall be provided for the supply to each unit. The collecting basin drain shall be connected to a discharge drain pipeline discharging to a nearby drain as approved by the S.O.. 'Flip' type cover shall be provided on the spray nozzles to protect against dust.
- (f) The appropriate safe condition signs shall be fixed in conspicuous places.

G4.9.3 Self-Contained Breathing Apparatus

- (a) Self-contained breathing apparatus shall provide complete respiratory protection from the surrounding atmosphere for a period of 60 minutes at normal breathing rate. The apparatus shall comply with BS EN 137 and shall be of a type approved by the UK Health and Safety Executive and the appropriate authority in Singapore.
- (b) Each set of apparatus shall incorporate an air cylinder, pressure reducing valve, pressure gauge, demand valve, panoramic vision face mask, bypass valve and adjustable harness with waist strap. The apparatus shall include a warning whistle to indicate approaching exhaustion of the cylinder.
- (c) A spare cylinder shall be supplied with each set.
- (d) All air cylinders shall be pre-charged with air meeting the requirements of BS EN 529.
- (e) Each set of apparatus shall be capable of accepting an additional face mask complete with its own demand valve and supply hose.
- (f) Each set shall be accommodated in a purpose designed wall mounting cabinet of a non-locking type with lift off front cover. The cabinet shall be suitably labelled. Storage racks shall also be provided to store empty cylinders.

G4.9.4 Cartridge Type Respirators

- (a) Cartridge (or canister) type respirators shall comply with BS EN 136 and shall be of the full-face piece type unless otherwise specified.
- (b) Cartridges (or canisters) shall comply with the relevant sections of BS EN 14387 and BS EN 143.
- (c) Each respirator shall be supplied with a pack of at least three spare cartridges or canisters.
- (d) Each set shall be accommodated in a purpose designed wall mounting cabinet of a non-locking type or, where specified, in a suitable hand carrying case. Cabinets or cases shall be suitably labelled. The labels shall state the intended use and maximum concentration of the contaminant and exposure time for which the cartridge or canister is suitable.

G4.9.5 Portable Gas Detectors

- (a) Portable gas detectors shall be suitable for the detection of flammable gas/vapour, toxic gas/vapour and/or oxygen deficiency as appropriate to the Works.
- (b) Flammable gas/vapour detectors shall comply with the relevant parts of BS EN 60079-29-1.
- (c) Detectors which are electrically operated shall be intrinsically safe to BS EN 60079-0 and BS EN 60079-11 and shall have the appropriate BASEEFA approval code but not less than Ex ia IIC T3.
- (d) Battery operated detectors which are re-chargeable shall be supplied with a charger suitable for mains operation at 240V A.C. unless otherwise specified.

- (e) Each detector shall have the following features:
 - Up to 3 different channels selection for different gases;
 - Selector switch OFF TEST ON;
 - Indicator lamp for POWER on;
 - Indicator lamp for ALARM condition;
 - Audible alarm;
 - Pushbutton for alarm RESET;
 - Analogue or digital indicator for GAS CONCENTRATION;
 - Carrying strap and/or belt clip;
 - Protective storage case; and
 - Spare battery.
- (f) Flammable gas detectors shall read % LEL and shall have the calibration gas engraved on a permanently fixed plate. The calibration gas shall be methane unless otherwise specified.
- (g) Toxic gas detectors shall read ppm of the specific toxic gas which shall be engraved on a permanently fixed plate.
- (h) Oxygen detectors shall read % O₂ up to at least 25% O₂. In situations where oxygen enrichment of the atmosphere may occur, for example by leakage from oxygen handling plant, two adjustable alarm levels shall be provided at nominal values of 18 % and 24 % respectively.

G4.9.6 First Aid Kits

- (a) First aid kits shall be contained in wall mounting boxes designed to protect the contents from dampness and dust and marked with a white cross on a green background.
- (b) Unless otherwise specified, each box shall contain the following items:
 - First aid guidance card;
 - 20 individually wrapped sterile adhesive dressings of assorted sizes;
 - 2 sterile eye pads with attachments;
 - 6 individually wrapped triangular bandages;
 - 6 safety pins;
 - 6 medium-sized individually wrapped sterile unmediated wound dressings (approximately 100 x 80 mm);
 - 2 large individually wrapped sterile unmediated wound dressings (approximately 130 x 90 mm);
 - 3 extra-large individually wrapped sterile unmediated wound dressings (approximately 280 x 175 mm);

Additionally, where eye wash bottle sets are not provided separately in the immediate vicinity.

- 3 No. 300 ml of sterile water or sterile normal saline solution (0.9%) in sealed disposable containers;

Additionally, where soap and water of drinking quality are not available in the immediate vicinity.

- 12 individually wrapped moist cleansing wipes;
- Scissors;
- Surgical tapes;
- Plaster Strips/band-aids;
- Antibiotics ointment; and
- Torchlight.

G4.9.7 Fire Blankets

- (a) Fire blankets shall be of the heavy duty industrial type complying with BS EN 1869, BS 7944 and contained in wall mounting cabinets suitably labelled with white lettering on a red background. The fire blankets shall be of the following characteristics and features:

- Non-asbestos;
- Non-reusable;
- Able to be used to cover small fires or wrap around the body; and
- Approximately of 1.8 m width x 1.8 m length.

G4.9.8 Ventilator Resuscitators

- (a) Ventilator resuscitators shall comply with BS EN ISO 10651-4 and shall be contained in a carrying case suitably labelled with white lettering on a green background.

G4.9.9 Automated External Defibrillator (AED)

- (a) Statutory approved AED in a dedicated kit shall be provided.

G4.9.10 Emergency Response Team (ERT) Rescue Equipment

- (a) A complete set of statutory approved ERT Rescue Equipment (including full body harness, tripod, rope, stretcher, ladder, etc.) shall be provided.

G4.9.11 Safety Signs

- (a) All labels and signs providing health and safety information or instructions shall comply with Singapore Standard SS508 Graphical symbols – safety colours and safety signs and / or BS ISO 7010 part 3. Fire safety signs shall comply with BS ISO 3864 part 1.
- (b) All labels and signs providing hazard identification shall be designed to be viewable from a distance of 30 m and comply with the Singapore Standard SS586 series of Standards GHS information and colour coding.

- (c) Warning signs to warn persons of the existence of confined space and its potential hazards shall be displayed at or near every access point to a confined space and comply with the Singapore Standard SS568 Code of practice for confined space.
- (d) Hazardous Areas according to the Hazardous areas classification assignment.
- (e) Signs which are not integral parts of the safety equipment and which are fixed separately to cabinets or to nearby surfaces shall be manufactured in accordance with the General Specification on Workmanship Requirements.

END OF SECTION